

Deployment Options and Cost Comparison

CoDesign Compass – Hosting & Deployment Recommendation

1. Purpose of This Document

This document explains the available deployment and hosting options for the CoDesign Compass system.

The purpose is to help the client understand:

- Where the system will be hosted
- How much the deployment may cost
- The advantages and disadvantages of each option
- Which option is recommended for the current project stage
- Additional operational costs such as domain names and database hosting

This comparison focuses on balancing:

Consideration	Description
Cost	Monthly and yearly operational cost
Stability	Reliability and uptime
Performance	Speed and responsiveness
Ease of Maintenance	How easy the system is to manage
Scalability	Ability to support more users later
Technical Complexity	Difficulty of deployment and maintenance

2. System Components That Need Hosting

The CoDesign Compass system contains several parts that must be deployed online.

Component	Purpose
Frontend Website	User interface that participants interact with
Backend API Server	Handles business logic and AI processing
Database	Stores submissions, profiles, and reports
AI Processing Services	Generates summaries and AI-assisted analysis
File Storage / Export	Stores generated reports and exports
Domain Name	Provides a public website URL

3. Current Technical Stack

Layer	Technology
Frontend	React + Vite
Backend	Spring Boot (Java)
Database	PostgreSQL
AI Integration	Ollama / OpenAI-compatible API
Deployment	Render / Netlify / Railway
Version Control	GitHub

4. Estimated System Size and Usage

The current system is designed for a small-to-medium user base.

Estimated Database Size

Data Type	Estimated Usage
User Profiles	Low
Survey / Issue Responses	Medium
AI Reports	Medium
Logs and Metadata	Low
Estimated Initial Database Size	Under 1 GB

Expected Growth

Time Period	Estimated Database Size
Initial Deployment	0.5 – 1 GB
After 1 Year	1 – 5 GB
Future Scaling	Can be upgraded if needed

At the current project stage, a small managed PostgreSQL database is sufficient.

5. Deployment Options

Option 1 – Keep the Current Setup

Overview

Continue using the existing deployment configuration:

- Frontend hosted separately
- Backend hosted separately
- Free or low-cost infrastructure

Typical Setup

Component	Platform
Frontend	Netlify
Backend	Render
Database	Render PostgreSQL

Estimated Cost

Service	Estimated Monthly Cost
Frontend Hosting	Free
Backend Hosting	Free – \$7 AUD
Database	Free
Domain Name	Optional
Total	\$0 – \$7 AUD/month

Advantages

- Lowest cost
- Already working
- No migration required
- Easy for students and small teams

Disadvantages

- Free services may “sleep” after inactivity
- First request may be slower
- Lower long-term reliability

Technical Notes

Category	Assessment
Stability	Moderate
Performance	Acceptable
Maintenance	Easy
Scalability	Limited
Recommended for Production	No

Summary

This option is suitable for prototypes and demonstrations, but less suitable for long-term client usage.

6. Option 2 – Railway Hosting

Overview

Host the entire application using Railway.

Typical Setup

Component	Platform
Frontend	Railway
Backend	Railway
Database	Railway PostgreSQL

Estimated Cost

Service	Estimated Monthly Cost
Hosting	\$0 – \$15 AUD
Database	Included / usage-based
Domain Name	Extra
Total	Approximately \$0 – \$15 AUD/month

Advantages

- Everything managed in one place
- Easy deployment workflow
- Fast performance
- No cold-start delay

Disadvantages

- Usage-based pricing may increase unexpectedly
- Less predictable monthly billing
- Smaller free-tier limitations

Technical Notes

Category	Assessment
Stability	Good
Performance	Good
Maintenance	Easy
Scalability	Good
Cost Predictability	Medium

Summary

Railway provides a simple deployment experience but may become more expensive as usage increases.

7. Option 3 – Render Only (Recommended)

Overview

Deploy the entire system using Render.

This means:

- Frontend hosted on Render

- Backend hosted on Render
- PostgreSQL database hosted on Render
- Single deployment platform for the whole system

This is the option selected for the project.

Proposed Architecture

Component	Deployment Method
Frontend	Render Static Site
Backend API	Render Web Service
Database	Render PostgreSQL
AI Service	External API / Local AI Integration
Domain Name	Custom .com domain

Estimated Monthly Cost

Service	Estimated Cost
Frontend Hosting	Free – Included
Backend Web Service	~\$7 AUD/month
PostgreSQL Database	~\$7 AUD/month
Domain Name (.com)	~\$15–16 AUD/year
SSL Certificate	Included
Total Estimated Monthly Cost	~\$14 AUD/month

Estimated Annual Cost

Item	Estimated Yearly Cost
Render Hosting	~\$168 AUD/year
Domain Name (.com)	~\$15–16 AUD/year
Total	~\$183–184 AUD/year

Recommended Domain Name

A professional `.com` domain is recommended because:

- Most familiar for users
- Easier to remember
- Looks more professional
- Better for future public deployment

Suggested Domain

Domain	Estimated Yearly Cost
codesigncompass.com	Approximately \$15–16 AUD/year

Advantages

Benefit	Explanation
Stable Hosting	No sleeping/free-tier delay
Predictable Cost	Monthly pricing remains stable
Simplified Maintenance	One platform for all services
Easier Troubleshooting	Centralized logs and deployments
Good Client Experience	Faster and more reliable
Easy Future Scaling	Services can be upgraded later

Disadvantages

Limitation	Explanation
Higher Cost	More expensive than fully free hosting
Vendor Dependence	System relies mainly on Render

Technical Assessment

Category	Assessment
Stability	Very Good
Performance	Very Good
Maintenance	Easy
Scalability	Good
Cost Predictability	High
Recommended for Client Deployment	Yes

Why This Option Was Selected

The tutor recommended:

- Clearer deployment structure
- More stable infrastructure
- Better explanation of operational costs
- A recommended deployment approach

After comparing all options, Render-only deployment was selected because it provides:

- A balance between cost and reliability
- Simple deployment and maintenance
- Stable performance for client usage
- Lower technical complexity
- Predictable long-term cost

This option is considered the most practical solution for the current project scope.

8. Option 4 – AWS / Google Cloud Platform

Overview

Use enterprise cloud providers such as:

- Amazon Web Services (AWS)
 - Google Cloud Platform (GCP)
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Estimated Cost

Service	Estimated Monthly Cost
Compute Services	\$20–60 AUD
Managed Database	\$10–30 AUD
Storage & Networking	Additional
Domain Name	Additional
Total	~\$20–100+ AUD/month

Advantages

Benefit	Explanation
Enterprise-Level Infrastructure	Used by large companies
High Scalability	Can support large user growth
Advanced Services	Many additional cloud features
High Reliability	Strong uptime guarantees

Disadvantages

Limitation	Explanation
Complex Setup	Requires advanced DevOps knowledge
Higher Maintenance	More configuration required
Expensive	Not suitable for small projects
Over-Engineered	Beyond current project requirements

Technical Assessment

Category	Assessment
Stability	Excellent
Performance	Excellent
Maintenance	Difficult
Scalability	Excellent
Cost Predictability	Medium
Recommended for Current Project	No

9. Overall Comparison

Option	Estimated Monthly Cost	Ease of Use	Stability	Scalability	Risk Level	Recommended
Current Setup	\$0-7 AUD	Very Easy	Moderate	Limited	Low	No
Railway	\$0-15 AUD	Easy	Good	Good	Medium	Possible
Render Only	~\$14 AUD	Easy	Very Good	Good	Low	Yes
AWS / GCP	\$20-100+ AUD	Difficult	Excellent	Excellent	High	No

10. Final Recommendation

Recommended Deployment Strategy

Selected Option:

Option 3 – Render Only

Recommendation Summary

Reason	Explanation
Stable Performance	Reliable hosting without free-tier sleeping
Predictable Cost	Approximately \$14 AUD/month
Easier Maintenance	Single platform for frontend, backend, and database
Better Client Experience	Faster loading and more reliable access
Appropriate Complexity	Suitable for the current project size
Future Scalability	Can upgrade resources later if needed

Recommended Production Setup

Component	Recommended Service
Frontend	Render Static Site
Backend	Render Web Service
Database	Render PostgreSQL
Domain Name	codesigncompass.com
SSL	Included with Render

11. Conclusion

The Render-only deployment solution provides the best balance between:

- Cost
- Stability
- Ease of maintenance
- Technical simplicity
- Future scalability

Although it is slightly more expensive than the free deployment setup, it offers a much more professional and reliable experience for client-facing deployment.

For the current stage of the CoDesign Compass project, this option is considered the most suitable long-term deployment strategy.